

CHARCUBE · DESIGN DECISIONS REGISTER

CharCube design decisions register

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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Source	docs/06-design-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.4	2026-10-02	Amish Chadha	Approved follow-ups carried out: every part priced; value-engineering result restated
0.3	2026-10-02	Amish Chadha	Amish approved the recommendations for all seven open decisions (2026-10-02); CCB-DDR-003 accepted; moved to decisions made; value-engineering note on the decided additions
0.2	2026-10-01	Amish Chadha	Budget treated as a value-engineering target
0.1	2026-09-30	Amish Chadha	Register opened with the open decisions from the review note, the decision records and the build plan work

CharCube design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/`; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`, CCB-BLD-001) describes the design as it stands and does not list open decisions.

■ Open decisions

None. All open decisions were decided on 2026-10-02.

■ To confirm when parts are bought

#	What to confirm	Why it matters	Source
1	The two drums' actual diameters and heights, and the gap between them (at least 40 mm)	Drum sizes vary by maker; the gas gap, brick positions and probe slot height depend on them	CCB-DDR-003
2	The 160 mm pipe's actual outside diameter and the 168 mm sleeve's bore	The collar is rolled to the pipe; the slip joints need about 2 mm clearance	CCB-DDR-003, P2 and P8
3	The effect of the 150 mm lid hole on draft (8 % less area than the throat bore), confirmed at the first burn (decided 2026-10-02)	The draft calculation does not model the lid entry; margin 2.4 at peak	CCB-DDR-003, P2; CCB-CAL-001 section 6
4	Probe lengths (500 mm and 300 mm) and sheath rating	The core probe must reach the retort axis from outside the blanket	CCB-DDR-003, P12
5	Blanket roll widths (681 mm drum strip, 540 mm jacket strip)	Sets how many joins each wrap has	Build plan sections 3.4, 3.14
6	Prices of every BOM line once quoted, including the parts added for construction	The estimated cost against the value-engineering target rests on indicative prices	CCB-CAL-001 section 11
7	The factor of 3 on gas-side heat transfer and the four velocity heads of pressure loss assumed for the spiral baffle	Heat to water (R6) and draft depend on them; they are assumptions until measured at TRL 4	CCB-CAL-001 sections 1, 7

■ Value engineering

Value-engineering target: USD 320. Estimated cost of the constructable design: USD 546 (USD 226 over the target).

Every BOM line is priced, with its basis in the line's notes (2026-10-02). The parts added for construction add USD 17 to the earlier USD 319; the additions decided on 2026-10-02 add USD 186: the fibre board USD 28, the block thermocouple USD 10, the hot-surface labels USD 8 and the lifting aid USD 160 (CCB-CAL-001 section 11).

- **Main cost drivers:** the lifting aid (USD 160), the welded water jacket with its spiral baffle insert, the logger, the plinth with its fibre board, the ceramic fibre blankets and the two drums (see `bom/bom.csv`).
- **Savings worth trying:** sharing one lifting aid among several kilns on a site, or hanging the winch from an existing strong beam or tree where there is one (most of USD 160); leaving the fibre board off later builds if a measured

burn without it keeps the blocks under about 150 °C (USD 28, as decided); a cheaper logger box; a clay and ash render in place of the side blanket (about USD 25, from the TRL 2 review); and re-pricing the small parts at purchase.

■ Decisions made

Date	Decision	Decided by	Record
2026-09-25	TRL 2 review items 1 to 8 and 10: open-vented annular water jacket; \$300 kiln budget with the safety kit separate; nested-drum retort; bundled straw and stalks first; burner throat with preheated air; logger as standard; ceramic fibre blanket; tripod-carried jacket; keep the name	Amish: "proceed with all of your recommendations across all batches. Make sure we don't proceed to TRL 4 on any of them."	CCB-DDR-001
2026-09-25	Items 11 to 16: air shroud in place of the 25 mm ring; spiral baffle insert and jacket blanket; lid and top band blanket; R5 relaxed to 5 h; R9 restated to 700 °C; safety kit stated in R10 and the BOM notes	Amish: "i accept all your recommendations, go with them across all repos."	CCB-DDR-002
2026-09-26	Item 17: budget top-up from \$300 to \$320 for kiln parts	Amish: "I am ok with the budget top ups."	CCB-DDR-002, Table 1a
2026-09-25	TRL 4 on hold	Amish	project.yaml ; CCB-DDR-001
2026-09-30	Make the design physically buildable while drawing the build plan; outstanding decisions kept in this register, not in the build plan	Amish: "If you are realising that the design cannot be built as per concept - fix the design assumptions to match and be physically feasible as you draw the illustrations."	CCB-DDR-003 (accepted on 2026-10-02, below)
2026-10-02	Design for construction accepted: the twelve changes of Table 1 and their knock-on changes, as made; the first burn confirms that the 150 mm lid hole (8 % less area than the throat bore) does not limit draft	Amish: "i approve your recommendations for all 555 open decisions."	CCB-DDR-003, Table 1
2026-10-02	Plinth: a 25 mm ceramic fibre board between the pan and the blocks from the first burn, with a thermocouple on a block; the board is dropped on later builds only if a measured burn without it keeps the blocks under about 150 °C	Amish: "i approve your recommendations for all 555 open decisions."	CCB-DDR-003, A2
2026-10-02	Jacket sleeve: the welder flares the bottom 20 mm of the sleeve to about 190 mm so it guides itself onto the throat	Amish: "i approve your recommendations for all 555 open decisions."	CCB-DDR-003, A3
2026-10-02	Heat-recovery unit: a lifting aid, a swinging arm on a post or a hand winch, is designed for the first build; users are asked through the partner how it should work, not whether it is needed	Amish: "i approve your recommendations for all 555 open decisions."	CCB-PRC-001, open questions
2026-10-02	First region, residue and partner: kept open under the portfolio rule, chosen by this rule: a region where rice or wheat straw is still burned in the open, with a partner that already runs residue or biochar trials. First candidate to approach: an	Amish: "i approve your recommendations for all 555 open decisions."	CCB-DDR-001 item 9; CCB-DDR-002 Table 2

Date	Decision	Decided by	Record
	agricultural university's extension service, for example Punjab Agricultural University in India		
2026-10-02	Rain cap: the conical cap is adopted in the model and drawings, on the three flat-bar legs and 60 mm gap set by the design for construction	Amish: "i approve your recommendations for all 555 open decisions."	docs/REVIEW.md , 2026-09-26, appearance item 1
2026-10-02	Hot-surface labels: durable ISO 7010 W017 hot-surface warning labels for the drum blanket and the jacket are added to BOM line 14	Amish: "i approve your recommendations for all 555 open decisions."	docs/REVIEW.md , 2026-09-26, appearance item 5